DËKKAL

Flood Risk Intelligence Platform

DATA PIPELINE & ML BACKEND

Technical Documentation — v1.0

April 2026

Field	Detail
Project	Dëkkal — Flood Risk Scoring for IARD Insurers
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Backend Port	8001
Status	Phase 1 Complete — Expert-based scoring, XGBoost pending

INTERNAL TECHNICAL DOCUMENT

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SECTION EXEC — EXECUTIVE SUMMARY

This document provides a complete technical account of the Dëkkal data pipeline and ML backend built during Phase 1. It covers six data integration tasks, the FastAPI scoring endpoint, and the current state of the scoring engine — including an honest assessment of its limitations and the roadmap to a trained XGBoost model.

STATUS	*Phase 1 delivers a fully functional rule-based scoring engine with all geospatial data sources connected. XGBoost training is the immediate next step — blocked only by dataset size, which Phase 2 will resolve.*

Phase 1 delivered:

- 6 data pipeline tasks completed — SoilGrids, ERA5-Land, CHIRPS P99, Multi-zone dataset, FastAPI /score endpoint, GPM IMERG
- 8 geospatial feature sources integrated into a single feature extraction pipeline
- Rule-based scoring engine (v1.3) with all features connected — elevation, slope, drainage, soil, P99, SAR, soil moisture
- FastAPI backend on port 8001 with geocoding, zone validation, and structured JSON output
- 40-sample multi-zone ground truth dataset (Dakar Centre, Pikine, Guédiawaye, Keur Massar)
- Final ML-ready dataset: dekkal_final_dataset_v1.csv — 10 annual samples × 8 features

SECTION 1 — SYSTEM ARCHITECTURE

1.1 Data Flow

The Dëkkal scoring pipeline follows a sequential feature extraction and aggregation architecture:

```
Input: Address or Coordinates (lat, lon)
         ↓
Geocoding (Nominatim / OpenStreetMap)
         ↓
Zone Validation (Dakar urban bbox check)
         ↓
Feature Extraction — 6 parallel sources:
  ├── Terrain (SRTM 30m via GEE)
  ├── Precipitation P99 (CHIRPS via GEE)
  ├── SAR Water Detection (Sentinel-1 via GEE)
  ├── Drainage Distance (OSM via osmnx)
  ├── Soil Properties (SoilGrids via GEE)
  └── Soil Moisture (ERA5-Land via GEE)
         ↓
Scoring Engine v1.3 (rule-based — XGBoost pending)
  ├── Historical Risk      (25%) ← SAR delta
  ├── Structural Vulnerability (35%) ← terrain + soil + drainage
  └── Extreme Scenario Risk   (40%) ← P99 ensemble
         ↓
JSON Response: score + components + explanations + confidence
```

1.2 Current Limitation — Rule-based vs ML

IMPORTANT

The scoring engine v1.3 uses manually defined weights (25/35/40%) and thresholds. These are expert-based rules, not learned parameters. XGBoost training is the next step — requires 100+ samples. Current dataset: 40 samples.

Approach	Status	Accuracy	Next Step
Rule-based scoring (v1.3)	Active — production	Expert-defined	Replace with XGBoost
XGBoost classifier	Pending	To be measured	After dataset expansion
LSTM nowcasting	Retired from v1.0	N/A	Deferred to v2.0

SECTION 2 — DATA PIPELINE — 6 TASKS

Task 1 — SoilGrids Soil Properties

Field	Value
Script	soil_gee.py
Output	dekkal_soil_gee.csv
Source	projects/soilgrids-isric (Google Earth Engine)
Resolution	250m
Status	Complete √
Note	SoilGrids REST API + OpenLandMap REST API both return null for West Africa — GEE is the only working source

Key Findings

Property	P10	P50 (Median)	P90	Interpretation
Clay 0-5cm	18.1%	20.5%	25.1%	Moderate impermeability — soil saturates under heavy rain
Sand 0-5cm	57.9%	64.7%	71.1%	Dominant texture — moderate natural drainage
Silt 0-5cm	10.0%	13.8%	18.5%	Secondary component

Insight: Dakar urban soil is sandy-loam (65% sand, 20% clay). Drains moderately in dry conditions but becomes impermeable when saturated — consistent with observed flood patterns.

Task 2 — ERA5-Land Antecedent Soil Moisture

Field	Value
Script	era5_dakar.py
Output	dekkal_era5_soil_moisture.csv
Source	ECMWF/ERA5_LAND/DAILY_AGGR (GEE)
Images	3,652 daily images (2015-2024)
Variable	volumetric_soil_water_layer_1 (0-7cm)
Status	Complete √

Key Findings

Group	Soil Moisture Mean	Precip Mean	Interpretation
flood=1 (2016-2020)	0.126 m³/m³	399mm	Lower moisture — counter-intuitive
flood=0 (2015,2021-2024)	0.153 m³/m³	603mm	Higher moisture — no flood

INSIGHT *Soil moisture alone does not predict floods in Dakar. Terrain features are the primary discriminating variables. This confirms Dëkkal's core thesis: flood risk requires multi-source geospatial analysis, not single-variable models.*

Task 3 — CHIRPS P99 Extreme Rainfall

Field	Value
Script	chirps_p99_dakar.py
Output	dekkal_chirps_p99_dakar.csv
Source	UCSB-CHG/CHIRPS/DAILY (GEE)
Images	6,885 (rainy season Jun-Oct, 1981-present)
Status	Complete √

Global Percentiles — Dakar Rainy Season

Percentile	mm/day	Interpretation
P50 (median)	0.0mm	Half of all days have zero rain (normal)
P90	9.99mm	Standard rainy day
P95	17.08mm	Heavy rain event
P99	34.0mm	Extreme event — exceeds drainage capacity

Drainage capacity proxy: 25mm/day for urban Dakar sealed surfaces. P99 (34mm) consistently exceeds this threshold — confirming structural flood risk.

Task 4 — Multi-Zone Ground Truth Dataset

Field	Value
Script	dekkal_dataset_zones.py
Output	dekkal_dataset_zones.csv

Field	Value
Zones	Dakar Centre, Pikine, Guédiawaye, Keur Massar
Samples	40 (4 zones × 10 years)
Balance	flood=1: 25 **|** flood=0: 15
Status	Complete √

Results by Zone

Zone	flood=1	flood=0	Observation
Dakar Centre	8/10	2/10	Quasi-systematically flooded
Guédiawaye	8/10	2/10	Highly vulnerable — large SAR delta
Pikine	7/10	3/10	Vulnerable — variable delta
Keur Massar	2/10	8/10	Noisy SAR signal — bbox needs refinement

NOTE *Keur Massar bbox captures non-urban water bodies that inflate the dry baseline. A tighter polygon around the residential zone will improve accuracy in v1.1.*

Task 5 — FastAPI /score Endpoint

Field	Value
Script	backend/main.py + api/routers/score.py
Port	8001
Endpoint	POST /api/v1/score
Geocoder	Nominatim (OpenStreetMap) — free, no API key
Status	Complete √

API Input Specification

```
// Option 1 — Address
{ "address": "Pikine Technopole, Dakar" }

// Option 2 — Coordinates
{ "lat": 14.75, "lon": -17.39 }
```

API Output — Canonical Response

```
{
```

```
    "location": { "lat": 14.7421, "lon": -17.3891,
      "address_normalized": "Pikine, Dakar Region, Senegal" },
    "score": 37,
    "risk_level": "Moderate",
    "components": {
      "historical_risk": 5.0,
      "structural_vulnerability": 74.8,
      "extreme_scenario_risk": 24.9
    },
    "explanations": [
      { "factor": "Low elevation (2.69m)", "impact": "High" },
      { "factor": "P99 rainfall exceeds drainage capacity", "impact": "Medium" },
      { "factor": "Moderate clay content (20.5%)", "impact": "Medium" }
    ],
    "decision_support": {
      "action": "file_review",
      "label": "File review recommended"
    },
    "confidence": 1.0,
    "meta": {
      "model_version": "v1.0",
      "data_freshness": "2026-04",
      "processing_time_ms": 6646,
      "data_completeness": "high"
    }
  }
```


Zone Validation

The API rejects requests outside the Dakar urban coverage area (lat [14.6, 14.95], lon [-17.6, -17.1]) with a 400 error. Tested and confirmed for Abidjan, Côte d'Ivoire.

Task 6 — GPM IMERG Historical Analysis

Field	Value
Script	gpm_p99_dakar.py + finalize_dataset.py
Output	dekkal_gpm_p99_dakar.csv + dekkal_final_dataset_v1.csv
Source	NASA/GPM_L3/IMERG_V07 (GEE)
Images	205,623 (rainy season 2000-present, 30-min resolution)
Scope	Historical analysis only — no nowcasting (deferred to v2.0)
Status	Complete √

GPM vs CHIRPS P99 Comparison

Year	GPM P99 (mm/day)	CHIRPS P99 (mm/day)	Ensemble P99
2015	39.6	49.9	44.8
2016	34.4	38.8	36.6

Year	GPM P99 (mm/day)	CHIRPS P99 (mm/day)	Ensemble P99
2017	30.9	28.7	29.8
2018	21.5	26.4	24.0
2019	42.7	31.8	37.2
2020	42.5	36.3	39.4
2021	46.6	56.3	51.4
2022	53.2	39.6	46.4
2023	40.7	33.1	36.9
2024	39.3	30.1	34.7

The ensemble P99 (average of GPM and CHIRPS) is more robust than either source alone, as the two datasets use different retrieval algorithms and calibration approaches.

SECTION 3 — FEATURE ENGINEERING

3.1 Complete Feature Set

Feature	Source	Value Range	Connected to Score?
elevation_m	SRTM 30m (GEE)	2-13m in Dakar	√ structural_vulnerability
slope_deg	SRTM terrain (GEE)	0-5° in Dakar	√ structural_vulnerability
hand_proxy	Derived: 1/(elev+0.1)	0.08-0.37	Informational
drain_distance_m	OSM via osmnx	0-3000m	√ via drain_risk_score
drain_risk_score	Derived from distance	0-100	√ structural_vulnerability
clay_pct	SoilGrids via GEE	18-25%	√ structural_vulnerability
sand_pct	SoilGrids via GEE	58-71%	Informational
soil_moisture_current	ERA5-Land via GEE	0.04-0.18 m^3/m^3	√ structural_vulnerability
p95_mm_day	CHIRPS via GEE	11-29mm	√ extreme_scenario_risk
p99_mm_day	CHIRPS via GEE	21-56mm	√ extreme_scenario_risk
p99_ensemble	GPM+CHIRPS average	24-51mm	√ extreme_scenario_risk
sar_dry_km2	Sentinel-1 (GEE)	0-0.002 km^2	√ historical_risk
sar_wet_km2	Sentinel-1 (GEE)	0-0.002 km^2	√ historical_risk
sar_delta_km2	Derived: wet-dry	−0.001 to +0.5	√ historical_risk

3.2 OSM Drainage Network

Field	Value
Library	osmnx 2.1.0
Features	145 LineStrings (canals, drains, ditches)
CRS	EPSG:32628 (UTM Zone 28N) — metric distances
Cache	In-memory — loaded once at API startup
Risk Logic	< 50m: high (overflow risk) | 50-200m: moderate | 200-500m: low-moderate | >500m: high (no drainage)

3.3 Known Feature Limitations

- SRTM 30m averaged over 100m buffer: insufficient for address-level discrimination in flat urban zones (all Dakar zones score 8-13m elevation)
- SAR 500m buffer: small urban parcels return near-zero water area — delta is almost always <0.001 km^2

- CHIRPS 5km resolution: smooths over micro-precipitation variations within a neighborhood
- OSM drainage: only 145 features mapped in Dakar — significant under-representation of informal drainage

ROADMAP	*v1.1 will add: (1) HAND index from MERIT DEM — true height above nearest drainage, (2) Google Open Buildings — imperviousness ratio per parcel, (3) Copernicus DEM 10m — finer elevation discrimination.*

SECTION 4 — SCORING ENGINE v1.3

4.1 Architecture

The scoring engine uses a three-component weighted aggregation. All weights are manually defined (expert-based) and will be replaced by XGBoost learned weights in v1.1.

```
final_score = (
    0.25 * historical_risk           +
    0.35 * structural_vulnerability  +
    0.40 * extreme_scenario_risk
)
```

4.2 Component Formulas

Historical Risk (25%)

```
# Based on SAR water detection delta
if sar_delta_km2 <= 0:  return 5.0
return min(100, sar_delta_km2 / 0.3 * 100)
```

Structural Vulnerability (35%)

Four sub-components, each calibrated to the real SRTM data range observed in Dakar (8-13m elevation, 2-3° slope):

```
# Sub-weights within structural_vulnerability:
elev_score  = elev_norm  * 35   # elevation (ELEV range: 5-20m)
slope_score = slope_norm * 20   # slope     (SLOPE range: 1-5°)
drain_score = drain_norm * 25   # drainage distance
soil_score  = (clay_norm*0.6 + sm_norm*0.4) * 20  # soil

structural_vulnerability = elev + slope + drain + soil
```

Extreme Scenario Risk (40%)

```
DRAINAGE_CAPACITY = 25.0  # mm/day — urban Dakar sealed surfaces

p99_score = min(70, (p99_ensemble - 25) / 25 * 70)
p95_bonus = min(30, max(0, (p95_ensemble - 25) / 25 * 30))
extreme_risk = p99_score + p95_bonus
```

4.3 Risk Level Classification

Score	Risk Level	Decision Support Action
0-30	Low	Standard underwriting — no additional review
31-55	Moderate	File review recommended
56-75	High	Field verification advised
76-100	Extreme	Mandatory human decision required

4.4 Test Results

Location	Coordinates	Score	Risk	Confidence	Key Driver
Pikine bas-fond	14.7421, -17.3891	37	Moderate	1.0	Elevation 2.69m
Plateau Dakar	14.6928, -17.4441	33	Moderate	1.0	Best terrain in sample
Guédiawaye	14.7789, -17.3912	37	Moderate	0.83	Elevation + drainage

LIMITATION

Score discrimination is limited (33-37 range) due to SRTM 30m resolution. All Dakar urban zones have similar elevation (8-13m). Address-level discrimination requires HAND index and building footprint data — planned for v1.1.

SECTION 5 — FASTAPI BACKEND

5.1 Project Structure

```
~/dekkal/backend/
├── main.py                      # FastAPI app entry point
├── api/
│   ├── models/
│   │   └── schemas.py           # Pydantic input/output schemas
│   ├── routers/
│   │   └── score.py             # POST /api/v1/score router
│   └── services/
│       ├── feature_engine.py    # GEE feature extraction
│       ├── scoring_engine.py    # Rule-based scoring v1.3
│       ├── geocoder.py          # Nominatim geocoding
│       └── drainage_distance.py # OSM drainage distance
```

5.2 Key Dependencies

Package	Version	Purpose
fastapi	latest	Async API framework
uvicorn	latest	ASGI server
pydantic	latest	Input/output validation
earthengine-api	latest	GEE data access
geopy	latest	Nominatim geocoding
osmnx	2.1.0	OSM drainage network
geopandas	latest	Spatial distance computation
xgboost	latest	ML model (pending training)
scikit-learn	latest	LOO-CV validation

5.3 Performance

Operation	Time	Bottleneck	Fix
Terrain (GEE)	~2s	Single call	Parallel calls in v1.1
Precipitation P99 (GEE)	~2s	Large collection	Cache result per zone
SAR features (GEE)	~3s	Two collections	Cache dry baseline
OSM drainage	<0.1s	Cached in memory	Already optimized
Soil (GEE)	~2s	250m resolution	Cache per 1km grid
ERA5 (GEE)	~1s	Monthly mean	Cache per month

Operation	Time	Bottleneck	Fix
Total (cold)	~10s	Sequential calls	Redis cache → target <2s

SECTION 6 — DATASETS & FILES

6.1 Data Pipeline Files

File	Location	Content	Size
flood_ground_truth_dakar_2015_2024.csv	data-pipeline/	Annual SAR labels + ERA5 + CHIRPS + GPM features	10 rows × 17 cols
dekkal_dataset_zones.csv	data-pipeline/	Multi-zone ground truth (4 zones × 10 years)	40 rows × 8 cols
dekkal_final_dataset_v1.csv	data-pipeline/	ML-ready merged dataset with P99 ensemble	10 rows × 20 cols
dekkal_soil_gee.csv	data-pipeline/	SoilGrids clay/sand distribution	3 rows (depths)
dekkal_era5_soil_moisture.csv	data-pipeline/	ERA5 seasonal soil moisture 2015-2024	10 rows
dekkal_chirps_p99_dakar.csv	data-pipeline/	CHIRPS P95/P99 annual 2015-2024	10 rows
dekkal_gpm_p99_dakar.csv	data-pipeline/	GPM IMERG P95/P99 annual 2015-2024	10 rows
dekkal_terrain_grid_200pts.csv	data-pipeline/	SRTM terrain risk for 200 random Dakar points	200 rows
CRU_senegal_annuel...csv	data-pipeline/	CRU TS4.05 annual precipitation 1901-2020	12,960 rows

6.2 Backend Service Files

File	Purpose
api/services/feature_engine.py	All GEE + terrain feature extraction functions
api/services/scoring_engine.py	Rule-based scoring v1.3 — to be replaced by XGBoost
api/services/geocoder.py	Nominatim address geocoding with Dakar validation
api/services/drainage_distance.py	OSM drainage distance calculation with caching
api/routers/score.py	POST /api/v1/score endpoint with full pipeline
api/models/schemas.py	Pydantic schemas for input/output validation

SECTION 7 — NEXT STEPS — XGBOOST TRAINING

The scoring engine v1.3 uses manually coded rules. The immediate next phase replaces these with a trained XGBoost classifier. Here is the 4-step plan:

Step 1 — Feature Extraction for 40-Zone Dataset

Extract all 14 features (terrain, SAR, soil, P99, ERA5) for each of the 40 geographic points in dekkal_dataset_zones.csv. This creates an enriched training dataset with real feature values per zone per year.

Step 2 — Train XGBoost Classifier

```
from xgboost import XGBClassifier
from sklearn.model_selection import LeaveOneOut

model = XGBClassifier(n_estimators=100, max_depth=3,
                      learning_rate=0.1, eval_metric='logloss')

# LOO-CV on 40 samples
loo = LeaveOneOut()
for train_idx, test_idx in loo.split(X):
    model.fit(X.iloc[train_idx], y.iloc[train_idx])
    y_pred.append(model.predict(X.iloc[test_idx]))
```

Step 3 — Save and Load Model

```
import joblib

# Save
joblib.dump(model, 'dekkal_xgboost_v1.pkl')

# Load in FastAPI (startup event)
model = joblib.load('dekkal_xgboost_v1.pkl')
score = int(model.predict_proba([features])[0][1] * 100)
```

Step 4 — Replace scoring_engine.py

Remove manual weights and thresholds. Replace score_location() with model.predict_proba(). Keep explanation layer (SHAP values) for transparency.

Target Architecture v1.1

Component	v1.0 (current)	v1.1 (target)
Scoring	Rule-based (expert weights)	XGBoost trained on 40+ samples

Component	v1.0 (current)	v1.1 (target)
Feature importance	Manually defined	SHAP values — data-driven
Discrimination	Limited (all zones ~35)	To be measured post-training
Confidence	Based on feature completeness	Based on model probability
Performance	~10s (sequential GEE calls)	<2s (Redis cache)
Elevation data	SRTM 30m	MERIT DEM + HAND index

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